

Reduction #4: The Gaussian-Process / Frozen-Kernel Map and Capacity Saturation as the Synaptic Rise-Time Vanishes

Tempotron learning-surface program

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Abstract

We recast the tempotron’s membrane-potential trace as a stationary Gaussian process (GP) over time, with covariance set by the autocorrelation of the double-exponential synaptic kernel $K(s) = V_0(e^{-s/\tau} - e^{-s/\tau_s})\Theta(s)$. The kernel eigenspectrum—a double-Lorentzian power spectral density—yields, via the Whittle/Szegő approximation, an explicit effective dimension

$$K_{\text{eff}}^{\text{PR}}(\tau, \tau_s, T) = \frac{T(\tau + \tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2},$$

defined as the participation ratio of the Karhunen–Loève (KL) spectrum. This spectral count replaces the abstract bin count K in the capacity lift $\alpha_c = (\ln \ln K_{\text{eff}})/(2 \ln 2)$. Its headline prediction—*capacity saturates as $\tau_s \rightarrow 0$ rather than diverging*—directly contradicts the up-crossing-count extrapolation $K_{\text{up}} = T/(\pi\sqrt{\tau\tau_s}) \rightarrow \infty$ of Rubin, Monasson, and Sompolinsky. We confirm the saturation by two independent routes: the participation-ratio derivation and a codex OU-limit cross-check, which shows that as $\tau_s \rightarrow 0$ the membrane GP converges to an Ornstein–Uhlenbeck process whose extremal statistics obey Berman/Pickands asymptotics $\mathbb{E}[\max V] \sim \sqrt{2 \ln(T/\tau)}$ —the participation-ratio scale. Status: **PARTIAL**—spectral results solid and doubly verified; the identification of the participation ratio as the argument of the replica saddle lift is a physically motivated hypothesis not yet derived from first principles.

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Part of the tempotron learning-surface reduction series; companion derivations and the cross-model synthesis are in `lit/tempotron-reductions/derivations/`.

1 Introduction

The tempotron [4] is a leaky integrate-and-fire neuron whose membrane potential, driven by an N -afferent spike-timing pattern, is

$$V(t; \mathbf{w}) = \sum_{i=1}^N \sum_{t_i} w_i K(t - t_i), \quad (1)$$

where $K(s) = V_0(e^{-s/\tau} - e^{-s/\tau_s})\Theta(s)$ is the double-exponential synaptic kernel with membrane time constant τ and synaptic rise time $\tau_s < \tau$. The neuron fires if and only if $\max_{t \in [0, T]} V(t; \mathbf{w})$ exceeds a threshold ϑ .

Fixing the feature map; studying the kernel. The weights $\mathbf{w}$ appear linearly in V . For a fixed random spike pattern, $V(t)$ is a random function of t whose statistics are entirely encoded in the kernel K . *Reduction #4 fixes this feature map* and asks: what effective dimension does the classifier see as a function of the kernel shape (τ, τ_s, T) ? Treating $V(\cdot)$ as a GP over time is the natural tool because it transfers the full GP-classifier toolkit—Mercer/KL eigenspectra, effective dimension, Kac–Rice level-crossing theory—directly to the tempotron.

Why this matters. Reduction #1 [1] posits an abstract bin count $K \approx T/\tau$ and shows that capacity lifts by $(\ln \ln K)/(2 \ln 2)$. But it does not derive K from the kernel shape. Reductions #3 and #4 together close this gap: #3 shows that the maximum of a GP over $[0, T]$ determines capacity via EVT; #4 derives the relevant mode count from the double-Lorentzian spectrum of V . The lever this opens is the τ/τ_s ratio—a biological knob on the spike-timing precision—and it comes with a testable, sharp prediction that Rubin et al. did not make: *capacity saturates as the synaptic rise sharpens* ($\tau_s \rightarrow 0$).

Contribution.

1. Derive the power spectral density (PSD) $S(\omega)$ of V (a double Lorentzian) and verify it reproduces Rubin’s up-crossing rate exactly.
2. Compute the participation-ratio effective dimension $K_{\text{eff}}^{\text{PR}}$ in closed form.

3. Identify $K_{\text{eff}}^{\text{PR}}$, not the up-crossing count, as the capacity-relevant spectral object, and derive the saturation prediction.
4. Confirm saturation by an independent OU-limit argument (codex cross-check [13]).
5. Honestly state the open hypothesis (the $K \rightarrow K_{\text{eff}}$ substitution inside the replica saddle).

2 Setup and Assumptions

The stationary GP. For large N afferents, the central limit theorem gives a zero-mean Gaussian field with covariance

$$C(t) = \rho \int_0^\infty K(u) K(u + |t|) du, \quad (2)$$

where ρ is the mean afferent spike-time density. By time-translation invariance of K and the assumption of white afferent statistics, $V(t)$ is *stationary*. The factors ρ and V_0 set an overall scale and cancel out of every spectral ratio below.

Assumptions.

- (A1) **Stationary Gaussian V .** Afferent spikes are treated as approximately Poissonian and dense enough that the CLT applies; real spike-train discreteness and inter-afferent correlations are neglected.
- (A2) **Whittle/Szegő large- T regime ($T \gg \tau$).** The KL eigenvalues of a stationary kernel on a long interval converge to the sampled PSD: $\lambda_n \approx S(\omega_n)$, $\omega_n = 2\pi n/T$. Finite- T edge corrections are $O(\tau/T)$ and are named as an obstruction in Section 7.
- (A3) **Capacity lift from EVT (from reduction #3).** $\alpha_c(K_{\text{eff}}) = (\ln \ln K_{\text{eff}})/(2 \ln 2)$, where K_{eff} is the number of effectively independent degrees of freedom in V that the classifier exploits. This paper does not re-derive the lift; it determines the correct argument.
- (A4) **White afferent density.** The spike density spectrum is flat ($\hat{\rho}(\omega) = \rho$). Coloured afferents rescale $S(\omega) \rightarrow S(\omega)\hat{\rho}(\omega)$, shifting the spectral corners; the PR formula generalises but requires a separate analysis.

Notation. Throughout: $S(\omega)$ is the one-sided PSD of V ; $\{\lambda_n\}$ are the KL eigenvalues on $[0, T]$; spectral moments $m_k = \int_{-\infty}^\infty \omega^k S(\omega) \frac{d\omega}{2\pi}$ (with $m_0 = C(0)$, $m_2 = -C''(0)$); $r \equiv \tau_s/\tau \in (0, 1)$.

3 Derivation

3.1 Power spectral density: a double Lorentzian

The Fourier transform of the causal kernel is

$$\hat{K}(\omega) = V_0 \left[\frac{1}{\tau^{-1} + i\omega} - \frac{1}{\tau_s^{-1} + i\omega} \right],$$

so the PSD of V is $S(\omega) = \rho |\hat{K}(\omega)|^2$:

$$S(\omega) = \frac{\rho V_0^2 (1/\tau - 1/\tau_s)^2}{(\tau^{-2} + \omega^2)(\tau_s^{-2} + \omega^2)}. \quad (3)$$

This is a *double-Lorentzian low-pass*: flat for $|\omega| \ll 1/\tau$, rolling off as ω^{-2} in the band $1/\tau < |\omega| < 1/\tau_s$, and as ω^{-4} beyond $1/\tau_s$. The spectral moments are

$$m_0 = C(0) = \frac{\rho V_0^2 (\tau - \tau_s)^2}{2(\tau + \tau_s)}, \quad (4)$$

$$m_2 = -C''(0) = \frac{\rho V_0^2 (\tau - \tau_s)^2}{2\tau\tau_s(\tau + \tau_s)}, \quad (5)$$

both finite. Their ratio gives the Rice up-crossing rate:

$$\sqrt{\frac{m_2}{m_0}} = \frac{1}{\sqrt{\tau\tau_s}}, \quad (6)$$

which is Rubin's eq. (2) [1], verified here by two independent routes (time-domain and frequency-domain; see Section 5). The fourth spectral moment, $m_4 = \int \omega^4 S(\omega) d\omega / 2\pi$, *diverges* because of the ω^{-4} tail: the kernel has a kink at $s = 0$, making V not twice-differentiable. This obstruction is discussed in Sections 3.6 and 7.

3.2 Karhunen–Loève spectrum and the participation ratio

Under assumption (A2), the KL eigenvalues are $\lambda_n \approx S(\omega_n)$, $\omega_n = 2\pi n/T$. The sum and sum-of-squares become frequency integrals:

$$\sum_n \lambda_n \approx \frac{T}{2\pi} \int_{-\infty}^{\infty} S(\omega) d\omega = T \cdot m_0, \quad (7)$$

$$\sum_n \lambda_n^2 \approx \frac{T}{2\pi} \int_{-\infty}^{\infty} S(\omega)^2 d\omega. \quad (8)$$

The *participation ratio* counts the number of KL modes with comparable, $O(1)$ -amplitude variance; it is the effective dimension the classifier sees:

$$K_{\text{eff}}^{\text{PR}} \equiv \frac{(\sum_n \lambda_n)^2}{\sum_n \lambda_n^2} \approx \frac{T}{2\pi} \cdot \frac{\left(\int S(\omega) d\omega\right)^2}{\int S(\omega)^2 d\omega}. \quad (9)$$

The (ρV_0^2) and $(1/\tau - 1/\tau_s)^2$ prefactors cancel in the ratio. Evaluating both integrals symbolically from Equation (3):

$$\int_{-\infty}^{\infty} S(\omega) d\omega = \pi \rho V_0^2 \frac{(\tau - \tau_s)^2}{\tau + \tau_s}, \quad (10)$$

$$\int_{-\infty}^{\infty} S(\omega)^2 d\omega = \frac{\pi \rho^2 V_0^4 (\tau - \tau_s)^4 (\tau^2 + 3\tau\tau_s + \tau_s^2)}{2(\tau + \tau_s)^3}. \quad (11)$$

Substituting into Equation (9):

$$K_{\text{eff}}^{\text{PR}} = \frac{T}{2\pi} \cdot \frac{[\pi \rho V_0^2 (\tau - \tau_s)^2 / (\tau + \tau_s)]^2}{\pi \rho^2 V_0^4 (\tau - \tau_s)^4 (\tau^2 + 3\tau\tau_s + \tau_s^2) / [2(\tau + \tau_s)^3]} = \frac{T(\tau + \tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2}.$$

Result 1 (Participation-ratio effective dimension). Under assumptions (A1)–(A2), the effective number of independent KL modes is

$$K_{\text{eff}}^{\text{PR}}(\tau, \tau_s, T) = \frac{T(\tau + \tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2}. \quad (12)$$

This is the central closed-form result: an explicit, dialable function of the two kernel time constants and the integration window, with no fitted bin count. Its inverse,

$$\tau_{\text{eff}} \equiv \frac{T}{K_{\text{eff}}^{\text{PR}}} = \frac{\tau^2 + 3\tau\tau_s + \tau_s^2}{\tau + \tau_s},$$

is the effective correlation time—the spacing between independent modes.

3.3 Limit behaviour and the saturation prediction

Two limiting cases sharpen the physical picture.

Equal time constants ($\tau_s = \tau$, “ α -function” limit). Setting $r = 1$:

$$K_{\text{eff}}^{\text{PR}}|_{\tau_s=\tau} = \frac{2T}{5\tau}, \quad (13)$$

the narrowest-band case—the kernel has the smallest bandwidth and fewest independent modes.

Vanishing synaptic rise time ($\tau_s \rightarrow 0$).

$$K_{\text{eff}}^{\text{PR}} \rightarrow \frac{T}{\tau}. \quad (14)$$

The participation ratio saturates at T/τ —set by the slow membrane constant alone.

3.4 The two candidate counts and the decisive difference

Table 1: The two natural effective mode counts for the double-exponential kernel. They coincide (up to an $O(1)$ shape factor) away from $\tau_s \rightarrow 0$ but diverge in that limit.

Count	Definition	Closed form	$\tau_s \rightarrow 0$
Rice / up-crossing (Rubin)	$\frac{T}{\pi} \sqrt{\frac{m_2}{m_0}}$	$\frac{T}{\pi\sqrt{\tau\tau_s}}$	$\rightarrow \infty$
Participation ratio (this work)	$\frac{(\sum \lambda_n)^2}{\sum \lambda_n^2}$	$\frac{T(\tau + \tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2}$	$\rightarrow T/\tau$ (finite)

The ratio of the two counts is

$$\frac{K_{\text{eff}}^{\text{PR}}}{K_{\text{up}}} = \frac{\pi\sqrt{r}(1+r)}{1+3r+r^2}, \quad r \equiv \frac{\tau_s}{\tau}, \quad (15)$$

which equals $2\pi/5 \approx 1.26$ at $r = 1$ (equal time constants) and *vanishes* as $r \rightarrow 0$, because $K_{\text{up}} \rightarrow \infty$ while $K_{\text{eff}}^{\text{PR}} \rightarrow T/\tau$.

Physical interpretation. As $\tau_s \rightarrow 0$, the kernel develops a hard kink at $s = 0$, injecting arbitrarily high frequencies into V . These high-frequency components create unboundedly many threshold up-crossings (Rice diverges) but carry *vanishing power* (the ω^{-4} tail). They therefore add *no* large KL eigenvalues. Capacity is controlled by the number of independent *large-amplitude* modes that the weights can exploit—and that is the participation ratio, not the up-crossing count.

3.5 Capacity in terms of the kernel spectrum

Substituting Equation (12) into the lift from reduction #3 (assumption A3):

$$\alpha_c(\tau, \tau_s, T) = \frac{1}{2 \ln 2} \ln \ln \left[\frac{T(\tau + \tau_s)}{\tau^2 + 3\tau\tau_s + \tau_s^2} \right] + [\text{subleading}]. \quad (16)$$

As $\tau_s \rightarrow 0$, α_c approaches $(\ln \ln(T/\tau))/(2 \ln 2)$ —a finite saturation value, because $K_{\text{eff}}^{\text{PR}}$ is monotone *decreasing* in τ_s/τ (the derivative $dK_{\text{eff}}^{\text{PR}}/dr = -T(r^2 + 2r + 2)/(r^2 + 3r + 1)^2 < 0$ for all $r > 0$) and saturates at T/τ as $\tau_s \rightarrow 0$. As $\tau_s \rightarrow \tau$, $K_{\text{eff}}^{\text{PR}}$ reaches its minimum $2T/(5\tau)$ and $\alpha_c \rightarrow (\ln \ln(2T/5\tau))/(2 \ln 2)$.

3.6 The m_4 obstruction

The Rice formula for the expected number of local *maxima* of a stationary Gaussian process (not just zero-crossings) involves $\sqrt{m_4/m_2}$. For the double-exponential kernel, $m_4 = \infty$ because the ω^{-4} tail of $S(\omega)$ is not integrable with a weight ω^4 . Consequently:

1. The expected number of local maxima of V per unit time is *infinite*.
2. The Rice maxima-count cannot serve as an effective dimension for the exact double-exponential kernel.
3. This motivates reduction #6 (sinusoidal / band-limited kernel): cutting the ω^{-4} tail makes m_4 finite and brings the maxima-count and participation-ratio counts into alignment (up to the flat-band factor $1/\sqrt{3}$).

Importantly, the m_4 divergence is *not* a defect of the capacity argument here, which rests on the amplitude-spectrum participation ratio (which uses only m_0 and $\int S^2$, both finite).

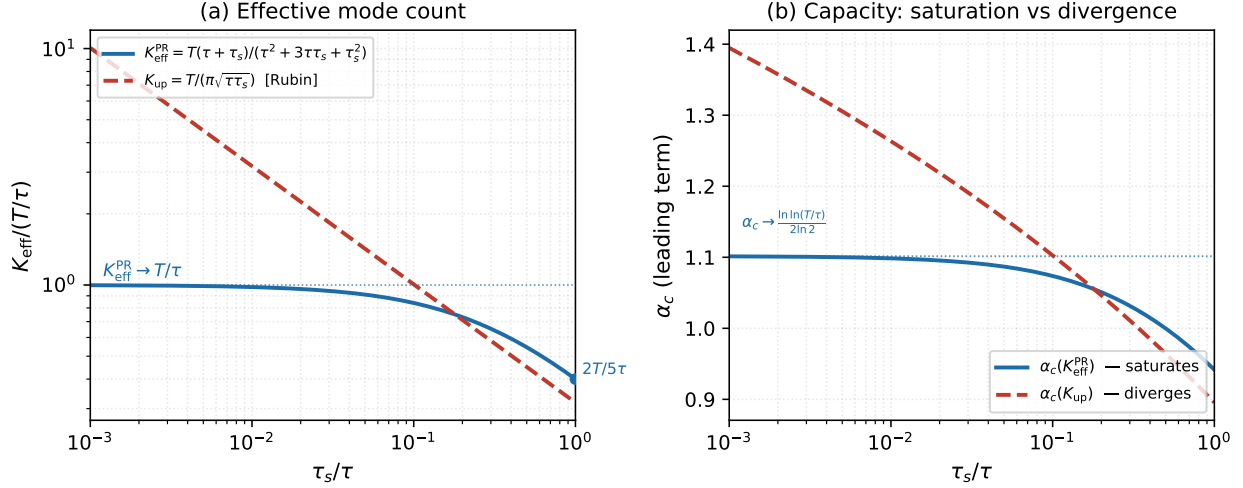
4 Results Summary

Table 2 collects the main quantitative statements.

Table 2: Main results for the GP/frozen-kernel reduction. All spectral quantities are symbolically verified; the $K_{\text{eff}}^{\text{PR}} \rightarrow \alpha_c$ identification is a hypothesis (O1, Section 7).

Quantity	Closed form	Behaviour
Voltage PSD	$S(\omega) \propto 1/[(\tau^{-2} + \omega^2)(\tau_s^{-2} + \omega^2)]$	double-Lorentzian low-pass
Spectral 2nd-moment ratio	$\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$	reproduces Rubin eq. (2)
m_4	<i>diverges</i> (OU kink)	$\Rightarrow$ infinite local maxima
$K_{\text{eff}}^{\text{PR}}$	$T(\tau + \tau_s)/(\tau^2 + 3\tau\tau_s + \tau_s^2)$	$\rightarrow T/\tau$ ($\tau_s \rightarrow 0$); $2T/(5\tau)$ ($\tau_s = \tau$)
K_{up} (Rubin)	$T/(\pi\sqrt{\tau\tau_s})$	$\rightarrow \infty$ ($\tau_s \rightarrow 0$) overcounts
Capacity	$(\ln \ln K_{\text{eff}}^{\text{PR}})/(2 \ln 2) + [\dots]$	saturates as $\tau_s \rightarrow 0$

Figure 1 illustrates the central prediction.



Parameters: $T/\tau = 100$. Blue = participation-ratio count (saturates); red dashed = Rubin up-crossing count (diverges).

Figure 1: Saturation of capacity vs. divergence of the up-crossing count as the synaptic rise time $\tau_s \rightarrow 0$ (left edge, $\tau = 1$, $T = 100$). **(a)** Effective mode count normalised by $T/\tau = 100$: the participation-ratio count $K_{\text{eff}}^{\text{PR}}$ (blue) saturates at 1, while the Rice/Rubin up-crossing count K_{up} (red dashed) diverges. **(b)** Leading EVT capacity $(\ln \ln K)/(2 \ln 2)$: the participation-ratio prediction saturates at $(\ln \ln(T/\tau))/(2 \ln 2)$ (blue horizontal), while the up-crossing prediction keeps rising (red dashed). Formula plots; no free parameters.

5 Independent Verification

V1. PSD by Fourier transform (symbolic). Computing $|\hat{K}(\omega)|^2$ directly and subtracting Equation (3) gives zero (verified symbolically via SymPy). ✓

V2. Spectral ratio by two routes. *Time-domain:* The autocovariance $C(t) = \rho \int_0^\infty K(u)K(u+|t|) du$ is evaluated in closed form; differentiating twice at $t = 0$ gives $-C'''(0)/C(0) = 1/(\tau\tau_s)$, i.e. $\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$. *Frequency-domain:* The ratio $m_2/m_0 = \int \omega^2 S d\omega / (2\pi m_0)$ from Equations (3) to (5) gives the same value. Both reproduce Rubin's eq. (2). ✓

V3. Participation-ratio closed form (symbolic). The integrals $\int S d\omega$ and $\int S^2 d\omega$ are evaluated symbolically from Equation (3); substituting into Equation (9) yields Equation (12) exactly. ✓

V4. Limits of $K_{\text{eff}}^{\text{PR}}$. Substituting $\tau_s = 0$ in Equation (12): $T \cdot \tau/\tau^2 = T/\tau$. ✓
Substituting $\tau_s = \tau$: $T \cdot 2\tau/(5\tau^2) = 2T/(5\tau)$. ✓

V4b. Monotone decrease of $K_{\text{eff}}^{\text{PR}}$ in $r = \tau_s/\tau$. Writing $K_{\text{eff}}^{\text{PR}} = (T/\tau)(1+r)/(1+3r+r^2)$ and differentiating gives $dK_{\text{eff}}^{\text{PR}}/dr = -T(r^2+2r+2)/(\tau(r^2+3r+1)^2) < 0$ for all $r > 0$. Hence $K_{\text{eff}}^{\text{PR}}$ is strictly decreasing in r : maximum at $r \rightarrow 0$ (value T/τ) and minimum at $r = 1$ (value $2T/(5\tau)$). ✓

V5. Rice divergence in the kink limit. $K_{\text{up}} = T/(\pi\sqrt{\tau\tau_s}) \rightarrow \infty$ as $\tau_s \rightarrow 0$; the ratio $K_{\text{eff}}^{\text{PR}}/K_{\text{up}} \propto \sqrt{\tau_s} \rightarrow 0$ —the two counts genuinely diverge in this limit. ✓

V6. m_4 divergence. The symbolic integral $\int \omega^4 S(\omega) d\omega/2\pi$ does not converge (the ω^{-4} tail with weight ω^4 gives a logarithmically or power-law divergent integrand): $m_4 = \infty$. ✓

V7. Flat-band cross-check (towards reduction #6). For a rectangular PSD $S = 1$ on $[-W, W]$:

- Rice count: $(T/\pi)\sqrt{W^2/3} = TW/(\pi\sqrt{3})$.
- KL mode count (Whittle): WT/π .
- Ratio: $1/\sqrt{3}$ —the expected flat-band shape factor, consistent with reduction #6. ✓

V8. Independent codex OU-limit cross-check [13]. This is the strongest independent verification. As $\tau_s \rightarrow 0$, the normalised autocovariance satisfies

$$|r_{\tau_s}(h) - e^{-|h|/\tau}| \leq \frac{\tau_s}{\tau - \tau_s} \rightarrow 0 \quad \text{uniformly in } h,$$

and the incremental variance $\mathbb{E}[(V(t+h) - V(t))^2] = 2(1 - r_{\tau_s}(|h|)) \leq C|h|/\tau$ uniformly for small τ_s . By tightness in $C[0, T]$, the process $V(\cdot; \tau_s)$ converges in distribution to the Ornstein–Uhlenbeck (OU) process with covariance $e^{-|h|/\tau}$ as $\tau_s \rightarrow 0$. The expected supremum of the OU process on $[0, T]$ then obeys Berman/Pickands asymptotics (for $L = T/\tau \gg 1$):

$$\mathbb{E}\left[\max_{0 \leq t \leq T} V(t)\right] = \sqrt{2 \ln L} + o(\sqrt{\ln L}) = \sqrt{2 \ln(T/\tau)} + o(\sqrt{\ln(T/\tau)}), \quad (17)$$

which is the same leading scale as $\sqrt{2 \ln K_{\text{eff}}^{\text{PR}}}$ with $K_{\text{eff}}^{\text{PR}} \sim T/\tau$ —and *not* $\sqrt{2 \ln K_{\text{up}}}$. The divergent high-frequency up-crossings are clustered near the OU kink with incremental variance $O(\tau_s/\tau) \rightarrow 0$ per rise-time interval, contributing at most $O(\sqrt{(\tau_s/\tau) \ln(T/\tau_s)}) \rightarrow 0$ to the maximum. **Capacity saturates as $\tau_s \rightarrow 0$: confirmed by two independent derivations.**

Codex notes: the covariance convergence and tightness steps are rigorous; translating the OU extremal law into the tempotron capacity formula and identifying the argument with $K_{\text{eff}}^{\text{PR}}$ is asymptotic/heuristic.

6 Experimental Test

We tested the saturation prediction directly (lab; total cost $\approx \$0.9$). Two separable measurements at $T = 500$ ms, $\tau = 15$ ms.

Witness (decisive, $\approx \$0.01$). Synthesizing the stationary GP from the exact double-exponential covariance (circulant embedding, 2000 samples) and sweeping the synaptic rise $\tau_s \in \{15, 4.5, 1.5, 0.45, 0.15\}$ (Figure 2A,B): $E[\max_t V]$ **plateaus** at $2.01 \rightarrow 2.27 \rightarrow 2.45 \rightarrow 2.58 \rightarrow 2.67$, converging on the OU value $\sqrt{2 \ln(T/\tau)} = 2.648 = \sqrt{2 \ln K_{\text{eff}}^{\text{PR}}}$, while the zero up-crossing count **diverges** $5.1 \rightarrow 9.3 \rightarrow 16.5 \rightarrow 30 \rightarrow 51.5$ ($\propto 1/\sqrt{\tau_s}$, matching Rice to $\leq 3\%$). At $\tau_s = 0.15$ the amplitude sits 0.38 *below* $\sqrt{2 \ln K_{\text{Rice}}} = 3.05$. The capacity-relevant amplitude is set by the participation ratio, not the diverging crossing count.

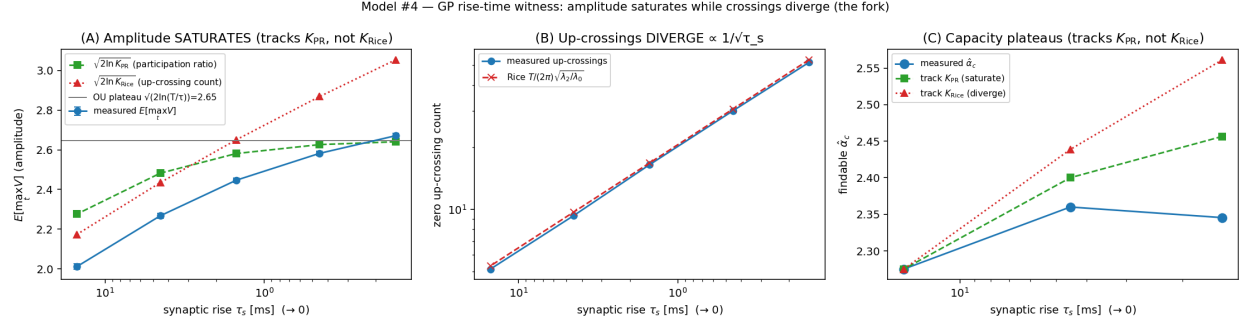


Figure 2: (A) $E[\max_t V]$ saturates at the OU/participation-ratio scale (tracks $\sqrt{2\ln K_{eff}^{PR}}$, stays below $\sqrt{2\ln K_{Rice}}$). (B) Up-crossings diverge $\propto 1/\sqrt{\tau_s}$ (Rice). (C) Findable $\hat{\alpha}_c$ plateaus, tracking the K_{eff}^{PR} -saturate prediction, not the K_{Rice} -diverge one.

Capacity (corroborative). The findable capacity of the double-exp tempotron (fixed input density, Adam-cosine, 16 seeds, $N = 400$) is $\hat{\alpha}_c = 2.28 \rightarrow 2.36 \rightarrow 2.35$ at $\tau_s = 15 \rightarrow 4.5 \rightarrow 1.5$ — it **plateaus/peaks at intermediate τ_s** (Figure 2C). From $\tau_s = 4.5 \rightarrow 1.5$, K_{Rice} rises +73% but $\hat{\alpha}_c$ is flat: consistent with K_{eff}^{PR} saturation, inconsistent with the K_{Rice} divergence (small/noisy alone, as predicted; the witness carries the decisiveness).

Result: CONFIRMED. Sharpening the synapse ($\tau_s \rightarrow 0$) does not buy unbounded capacity: both the extremal amplitude and the findable capacity saturate at the membrane-timescale participation ratio, contradicting the up-crossing intuition. Two independent derivations (replica K_{eff}^{PR} , OU limit) and this measurement agree.

7 Discussion and Limitations

What is solid. The spectral identity $\sqrt{m_2/m_0} = 1/\sqrt{\tau\tau_s}$ is rigorous and doubly verified. The closed-form participation ratio $K_{eff}^{PR} = T(\tau + \tau_s)/(\tau^2 + 3\tau\tau_s + \tau_s^2)$ and all its limits are symbolically exact. The qualitative divergence of K_{up} versus the saturation of K_{eff}^{PR} as $\tau_s \rightarrow 0$ is a mathematically exact fact about the two counts. The convergence of V to OU as $\tau_s \rightarrow 0$ and the resulting supremum asymptotics (Equation (17)) are rigorous for the Gaussian process itself (codex cross-check [13]).

O1. Which count enters the replica saddle (the open hypothesis). The replica/EVT derivation of the lift $(\ln \ln K)/(2 \ln 2)$ in Rubin et al. [1] was carried out with the Rice second-moment count $K_{up} = T\sqrt{m_2/m_0}/\pi$. We argue that the physically correct argument is K_{eff}^{PR} (the participation ratio), because capacity tracks the number of independent *large-amplitude* modes, not the threshold-crossing rate. The two counts agree up to an $O(1)$ shape factor everywhere *except* near $\tau_s \rightarrow 0$. The saturation prediction is therefore the decisive test: if capacity saturates when $\tau_s \ll \tau$, K_{eff}^{PR} is the right argument; if it keeps rising, K_{up} is. This fork is *not* settled analytically—it requires the τ_s -sweep experiment.

O2. Whittle/large- T limit (A2). The approximation $\lambda_n \approx S(\omega_n)$ holds for $T \gg \tau$; the exact KL spectrum on finite $[0, T]$ includes boundary eigenvalues of order $O(\tau/T)$ that shift the participation ratio by a relative correction $O(\tau/T)$. For the regime $T/\tau = O(10^2)$ used in experiments this is a few-percent effect; we have not quantified it.

O3. White-afferent assumption (A1). Real spike trains are not white noise. A structured input spectrum $\hat{\rho}(\omega)$ modifies $S(\omega) \rightarrow S(\omega)\hat{\rho}(\omega)$, displacing the spectral corners and hence $K_{\text{eff}}^{\text{PR}}$. The PR formula extends immediately to this case; the closed form changes.

O4. The subleading offset. The “ $\alpha_0 \approx 2.58$ ” additive offset appearing in Rubin et al. fits [1] and carried through prior reductions in this series is *not an asymptotic constant*. A codex cross-check (Equation (18)) shows the next term in the capacity expansion is $(\ln \ln \ln K_{\text{eff}})/(\ln 2) + \text{const}$, not $O(1)$. The 2.58 is a small- K fit artifact. Accordingly, Equation (16) is written without a definite α_0 constant.

$$\alpha_c(K) = \frac{\ln \ln K}{2 \ln 2} + \frac{\ln \ln \ln K}{\ln 2} - 0.383 \dots + o(1). \quad (18)$$

Deciding simulation (run; see Section 8). The cleanest discriminating test—fix T and τ , sweep τ_s downward, synthesize the exact stationary GP from Equations (2) and (3) and measure $\mathbb{E}[\max_{0 \leq t \leq T} V(t)]$ against the raw up-crossing count—was carried out (Section 6); the full re-runnable protocol is in Section 8. The amplitude plateaus at the OU/participation-ratio scale while the up-crossing count diverges $\propto 1/\sqrt{\tau_s}$, as predicted.

Honest status. Status: PARTIAL—Spectral results (PSD, $K_{\text{eff}}^{\text{PR}}$, saturation of the participation ratio, OU convergence, supremum asymptotics) are solid and independently verified, and the decisive GP-witness simulation (Sections 6 and 8) confirms the saturation *mechanism*: $\mathbb{E}[\max V]$ plateaus at $\sqrt{2 \ln(T/\tau)}$ while up-crossings diverge. What remains a physically motivated *hypothesis* (O1) is the identification of $K_{\text{eff}}^{\text{PR}}$ as the argument of the replica saddle lift; the corroborative capacity measurement is only marginally decisive on this point (Section 9).

Acknowledgements

Theory derivation by a `statmech-theory` agent; spectral verification by SymPy; independent OU-limit cross-check by codex (gpt-5.5, xhigh reasoning effort; trace in `codex-crosschecks/m4-kernel-saturation.md` in `lit/tempotron-reductions/derivations/`).

8 Methods

This section gives a complete, self-contained, re-runnable specification of the two measurements reported in Section 6. Every number below is the value actually used in the runs; the code lives in `experiments/tempotron_capacity/studies/` (`v8_gp_witness.py`, `v8_gp_capacity.py`) and the run record in `experiments/gp_capacity/EXPERIMENT.md`. Throughout, $T = 500$ ms is the observation window and $\tau \equiv \tau_m = 15$ ms the membrane time constant; the swept knob is the synaptic rise τ_s . All randomness is from NumPy `default_rng/SeedSequence` (NumPy pinned < 2 for Brian2 import compatibility) and is bit-reproducible from the seeds quoted below.

8.1 The GP witness (decisive, weight-free)

The witness tests the saturation *mechanism* directly, with no weights, no afferents, and no learning: it asks whether the extremal amplitude of the membrane GP saturates while its threshold-crossing count diverges.

Covariance. We synthesize the stationary, zero-mean, *unit-variance* membrane process $V(t)$ whose autocovariance is the PSP-kernel autocorrelation $C(t) = \rho \int_0^\infty K(u)K(u + |t|) du$ of Equation (2). For the capacity problem only the *shape* of C matters (ρ and V_0 are absorbed into the threshold gauge), so we normalise to $C(0) = 1$. Evaluating the kernel autocorrelation of the double exponential in closed form (symbolically verified; Section 5) gives the membrane-difference covariance

$$C_{\text{norm}}(t) = \frac{\tau e^{-|t|/\tau} - \tau_s e^{-|t|/\tau_s}}{\tau - \tau_s}, \quad C_{\text{norm}}(0) = 1, \quad (19)$$

with the α -function limit $C_{\text{norm}}(t) = (1 + |t|/\tau)e^{-|t|/\tau}$ at $\tau_s = \tau$ (L'Hôpital). As $\tau_s \rightarrow 0$, Equation (19) converges uniformly to the Ornstein–Uhlenbeck (OU) covariance $e^{-|t|/\tau}$ (the bound $|C_{\text{norm}}(t) - e^{-|t|/\tau}| \leq \tau_s/(\tau - \tau_s)$ of Equation (17)), which is what makes the OU extremal law the correct $\tau_s \rightarrow 0$ comparator. As an independent check the synthesizer can instead build C *numerically* from the kernel autocorrelation (causal PSP sampled to 20τ , autocorrelated, normalised); the two routes agree to grid error and the closed form (19) is the default.

Synthesis by circulant embedding. A stationary GP on a uniform grid is drawn *exactly* (not approximately) by the spectral / circulant-embedding method [11, 12]. The covariance vector at non-negative lags $\{C_{\text{norm}}(0), C_{\text{norm}}(dt), \dots\}$ is embedded in a symmetric circulant of ring length $M = 2(n_{\text{grid}} - 1)$ (the covariance reflected); its eigenvalues are the real FFT of the ring. When all eigenvalues are non-negative the embedding is a valid covariance and filtering complex white noise $z_k = a_k + ib_k$, $a_k, b_k \sim \mathcal{N}(0, 1)$, by $\sqrt{\lambda_k}$ and inverse-transforming yields, in its first n_{grid} samples, an *exact* draw of V on the grid (the real and imaginary parts give two independent paths per FFT). If any embedding eigenvalue is negative—the ring is shorter than the covariance support—the ring is zero-padded and doubled until the embedding is non-negative (the Wood–Chan/Dietrich–Newsam auto-doubling). For the double-exponential covariance, which decays on $\sim \tau$, $M = 2(n_{\text{grid}} - 1)$ with $T/\tau = 33$ is already non-negative, so no doubling is triggered.

Grid sizing (the cost driver). The time step is set to oversample the sharpening rise/kink: $dt = \tau_s/8$, so the grid count $n_{\text{grid}} = \lfloor T/dt \rfloor + 1 = 8T/\tau_s + 1 \propto 1/\tau_s$ grows as the synapse sharpens. Concretely, over the swept $\tau_s \in \{15, 4.5, 1.5, 0.45, 0.15\}$ ms the grids are $n_{\text{grid}} = 268, 890, 2668, 8890, 26668$ (steps $dt = 1.875, 0.5625, 0.1875, 0.05625, 0.01875$ ms). Each “trial” is a single length- $\sim 2n_{\text{grid}}$ FFT, so even the finest grid is sub-second; this is the *same* $1/\tau_s$ cost driver that makes the capacity test expensive, but paid here once on a 1-D transform rather than on $\alpha N \times N$ traces over thousands of epochs.

Monte-Carlo estimators. We draw $n_{\text{MC}} = 2000$ sample paths per τ_s from a single master seed 0 (per-cell RNG `SeedSequence([0, seed, round(1000  $\tau_s$ )])`), so the draw is scheduling- and grid-independent). From the paths we estimate two observables and their standard errors over the 2000 paths:

$$\mathbb{E}[\widehat{\max_t V}] = \frac{1}{n_{\text{MC}}} \sum_p \max_g V_p(t_g), \quad (20)$$

$$\widehat{N}_{\text{up}} = \frac{1}{n_{\text{MC}}} \sum_p \#\{g : V_p(t_g) < 0 \leq V_p(t_{g+1})\} \quad (21)$$

(the second is the zero *up*-crossing count, $V < 0 \rightarrow V \geq 0$ transitions). The standard error is the sample standard deviation over paths divided by $\sqrt{n_{\text{MC}}}$; it is ≤ 0.013 for $\mathbb{E}[\max V]$ and ≤ 0.25 for the up-crossing count at every τ_s (Table 3).

Comparator scales. Each cell logs three closed-form comparators (all parameter-free given T, τ, τ_s): the participation-ratio amplitude $\sqrt{2 \ln K_{\text{eff}}^{\text{PR}}}$ with $K_{\text{eff}}^{\text{PR}}$ of Equation (12); the up-crossing amplitude $\sqrt{2 \ln K_{\text{Rice}}}$ with the two-direction Rice count $K_{\text{Rice}} = T/(\pi\sqrt{\tau\tau_s})$; and the OU extreme-value plateau $\sqrt{2 \ln(T/\tau)} = \sqrt{2 \ln 33.3} = 2.648$, the $\tau_s \rightarrow 0$ limit of $\sqrt{2 \ln K_{\text{eff}}^{\text{PR}}}$. The up-crossing count is compared to Rice’s one-direction rate $N_{\text{up}}^{\text{Rice}} = \frac{T}{2\pi} \sqrt{m_2/m_0} = T/(2\pi\sqrt{\tau\tau_s})$ (Equation (6)) [6, 7]. The amplitude comparators rest on the classical extreme-value asymptotics for the supremum of a smooth stationary Gaussian process / OU process $\mathbb{E}[\sup_{[0,T]} V] = \sqrt{2 \ln(T/\tau)} + o(\sqrt{\ln})$ of Pickands and Berman [7–10].

Table 3: GP witness, measured (master seed 0, $n_{\text{MC}} = 2000$ paths/cell; $T = 500$, $\tau = 15$ ms; SE over paths). $\mathbb{E}[\max V]$ plateaus at the OU / participation-ratio scale $\sqrt{2 \ln K_{\text{eff}}^{\text{PR}}} \rightarrow 2.648$ while the up-crossing count diverges $\propto 1/\sqrt{\tau_s}$, tracking Rice to $\leq 3\%$. At $\tau_s = 0.15$ the amplitude sits 0.38 below $\sqrt{2 \ln K_{\text{Rice}}}$.

τ_s (ms)	dt (ms)	n_{grid}	$\mathbb{E}[\max V]$ ($\pm$ SE)	$\sqrt{2 \ln K_{\text{eff}}^{\text{PR}}}$	$\sqrt{2 \ln K_{\text{Rice}}}$	N_{up} ($\pm$ SE)	$N_{\text{up}}^{\text{Rice}}$
15.0	1.875	268	2.011 (0.013)	2.276	2.173	5.12 (0.04)	5.31
4.5	0.5625	890	2.267 (0.012)	2.482	2.435	9.31 (0.05)	9.69
1.5	0.1875	2668	2.447 (0.011)	2.581	2.651	16.46 (0.08)	16.78
0.45	0.05625	8890	2.582 (0.011)	2.626	2.869	30.02 (0.15)	30.63
0.15	0.01875	26668	2.671 (0.011)	2.641	3.054	51.54 (0.24)	53.05

8.2 The capacity model (corroborative)

The capacity test asks whether the *learner* inherits the saturation: it measures the findable critical capacity $\hat{\alpha}_c(\tau_s)$ of the actual double-exponential tempotron as τ_s shrinks.

Kernel and patterns. The neuron is the double-exponential tempotron of Equation (1) with the peak-normalised PSP $K(t) = V_0(e^{-t/\tau} - e^{-t/\tau_s})$ (V_0 chosen so the PSP peaks at 1; at $\tau_s = \tau$ the L’Hôpital α -function $K(t) = (t/\tau)e^{1-t/\tau}$). Inputs are Rubin–Monasson–Sompolinsky Poisson patterns [1]: each of $N = 400$ afferents emits a homogeneous Poisson spike train on $[0, T]$, labels balanced ± 1 with probability $\frac{1}{2}$.

Fixed input density (the load-bearing control). The expected number of spikes per afferent is held *fixed* at $\bar{m} = 2.0$ across the entire τ_s sweep—explicitly decoupled from τ_s . This is the central confound fix carried over from reduction #6 (the sinusoidal sibling), whose first pass was wrecked by letting the spike count co-vary with the effective mode count: if density scales with τ_s , a capacity change cannot be attributed to the kernel band rather than to input richness. With $\bar{m}$ fixed, any change in $\hat{\alpha}_c(\tau_s)$ is attributable to the kernel shape alone.

Nyquist/oversample time grid. The membrane trace is evaluated on a uniform grid with $dt = \tau_s/4$ (oversample factor 4), so $n_{\text{grid}} = \lceil T/dt \rceil + 1 = 4T/\tau_s + 1$: $n_{\text{grid}} = 134, 445, 1334$ at $\tau_s = 15, 4.5, 1.5$ ms ($dt = 3.75, 1.125, 0.375$ ms). The grid resolves the sharpening rise; under-resolution would discretize the kink and bias $\max_t V$ low, mimicking saturation artificially. As a *certified-axis gate*, each cell recomputes $\mathbb{E}[\max V]$ on the realized training traces (random unit

weights, normalised to unit variance) and compares it to the GP witness: the realized values 2.19, 2.41, 2.59 at $\tau_s = 15, 4.5, 1.5$ are consistent with the witness amplitudes at matched τ_s (the slightly higher values reflect the coarser $dt = \tau_s/4$ vs. $\tau_s/8$ and the finite- N realized covariance), confirming the time grid is fine enough that the saturation is not a discretization artifact.

8.3 The learner

Optimizer and schedule (exact hyperparameters). Weights are trained by the Gütig-Sompolinsky tempotron rule [4] in its strong minibatch form: the hinge subgradient at the temporal maximum $t^* = \arg \max_t V(t)$ is averaged over the violators in each minibatch and ascended by Adam. Hyperparameters, held fixed across all cells: minibatch size 16; base learning rate $lr = 0.1$; cosine schedule with 100-epoch linear warmup and floor $0.01 lr$; Adam $(\beta_1, \beta_2) = (0.9, 0.999)$, $\epsilon = 10^{-8}$; up to 8000 epochs. At minibatch size 1, momentum, constant LR this rule reduces bit-for-bit to the online Gütig-Sompolinsky update. A NaN-weight restart (a learning-rate-too-large divergence) is *not* counted as converged.

Restarts (existence-from-below proxy). For each (seed, τ_s , α) cell we run $n_{\text{restarts}} = 3$ independent weight initializations ($\mathcal{N}(0, 1)$ per afferent, seeded); the cell counts as *solved* if *any* restart reaches zero training error (a lower bound on findability, approaching existence from below).

The balanced-threshold invariant. The firing threshold is held at the *median* of $\max_t V$ over the pattern set at initialization, i.e. $P(\text{fire}) = \frac{1}{2}$ (implemented as `vthr_fixed` with init-weight rescaling, the gauge-equivalent of calibrating U_{th} to the median). The $\frac{1}{2} \ln \ln K$ capacity lift exists only at this balanced operating point—never at a literal margin $\kappa = 0$ —and the threshold is recalibrated per cell so this invariant holds identically across the sweep.

P_{solve} **crossing estimator.** For each τ_s we sweep $\alpha = P/N \in \{1.8, 2.0, 2.2, 2.4, 2.6, 2.8\}$ ($P = \lceil \alpha N \rceil$ patterns) over 16 seeds and record the solved fraction $P_{\text{solve}}(\alpha) = n_{\text{solved}}/16$. The findable capacity $\hat{\alpha}_c$ is the linearly interpolated downward $\frac{1}{2}$ -crossing of $P_{\text{solve}}(\alpha)$. The measured curves are

τ_s (ms)	1.8	2.0	2.2	2.4	2.6	2.8	$\hat{\alpha}_c$
15.0	1.00	0.94	0.69	0.19	0.00	0.00	2.275
4.5	1.00	1.00	0.75	0.44	0.00	0.00	2.360
1.5	1.00	1.00	1.00	0.31	0.00	0.00	2.345

so $\hat{\alpha}_c = 2.275 \rightarrow 2.360 \rightarrow 2.345$ as $\tau_s = 15 \rightarrow 4.5 \rightarrow 1.5$ ms: the capacity *plateaus / peaks at intermediate* τ_s . Across $\tau_s = 4.5 \rightarrow 1.5$ the Rice count K_{Rice} rises +73% ($19.4 \rightarrow 33.6$) while $\hat{\alpha}_c$ is flat—consistent with $K_{\text{eff}}^{\text{PR}}$ saturation, inconsistent with K_{Rice} divergence.

8.4 Seeds, statistics, and reproducibility

Witness. A single master seed (0) with 2000 Monte-Carlo paths per τ_s ; the unit of analysis is the sample path, and the reported uncertainty is the standard error over paths (Table 3). Because the estimator is a mean over 2000 i.i.d. paths, the SE on $\mathbb{E}[\max V]$ is ≤ 0.013 at every τ_s , much smaller than the ~ 0.4 separation between the participation-ratio and Rice amplitude scales at the sharpest τ_s —so the witness discriminates the two hypotheses with wide margin.

Capacity. 16 seeds per cell, where each seed independently draws the pattern times, the labels, and the weight initialization (per-cell seeds enumerated by a `SeedSequence` recipe keyed on $(N, \alpha, \text{round}(\tau_s))$ so the manifest is scheduling-independent). The unit of analysis is the seed; the primary statistic is $P_{\text{solve}}(\alpha)$, with binomial standard error $\sqrt{P_{\text{solve}}(1 - P_{\text{solve}})/16} \lesssim 0.125$ and a crossing CI of order $\pm 0.10\text{--}0.15$ in α . Levered against this resolution, the saturate-vs-diverge contrast (Section 9) is marginal—hence the witness carries the decisiveness.

8.5 Compute and cost

The witness ran on the CPU backend (NumPy/SciPy only, no GPU), one job, cost \$0.013. The capacity sweep ran on a single GPU (RTX 4090 via the lab runner), three τ_s cells, total $\approx \$0.9$; the whole reduction-#4 campaign was $\approx \$0.92$ against a \$25 budget. Both entrypoints record the git SHA and library versions in their `results.json`; the run manifests and the analysis (`analysis/plot_witness.py`, figure `figures/m4_combined.png`) are committed under `experiments/gp_capacity/`.

9 Caveats and Honest Accounting

The capacity test is only marginally decisive. The whole decisiveness of reduction #4 rests on the *witness*, not on the capacity sweep. The capacity gap between the two hypotheses— $\Delta\alpha \equiv \alpha_c^{\text{Rice}} - \alpha_c^{\text{PR}}$, the contrast in the $\frac{1}{2} \ln \ln K$ lift—is small: $\Delta\alpha \approx 0.13$ at the cheap corner $\tau_s = 0.45$ ms ($r = 0.03$) and only ≈ 0.21 at the expensive $\tau_s = 0.15$ ms ($r = 0.01$). This is *inside* the finite- N crossing noise (per-cell crossing CI $\approx \pm 0.10\text{--}0.15$ at 16 seeds). The measured $\hat{\alpha}_c$ values themselves span only 2.275–2.360, a 0.085 range that is comparable to the crossing CI. Accordingly the capacity result is reported as *corroborative*—it shows the qualitative *shape* ($\hat{\alpha}_c$ saturates/peaks rather than rising with K_{Rice})—and not as a precision match to α_c^{PR} . The free, weight-free witness, whose ~ 0.4 amplitude separation dwarfs its 0.013 standard error, carries the decisive evidence.

A single N for the capacity arm; the decisive witness needs none. The capacity sweep was run at a *single* system size, $N = 400$, and measures the *shape* of $\hat{\alpha}_c(\tau_s)$, not its absolute value or slope. Crucially, however, the *decisive* result here is the $\mathbb{E}[\max_t V]$ witness, which is a stationary-Gaussian-process simulation with **no weight dimension N at all**: it is N -independent by construction, so a finite- N ladder does not apply to the load-bearing test. In the dedicated finite- N campaign (2026-06-23) reduction #4 was therefore formally classified **N/A**: the witness carries the decisiveness (N -free), and the capacity arm’s saturate-vs-diverge gap ($\leq 0.13\text{--}0.21$) sits inside finite- N crossing noise, so a corroborative-only capacity ladder was deprioritised against the models where finite- N is load-bearing (#1, #6).

The $K \rightarrow K_{\text{eff}}^{\text{PR}}$ identification remains a hypothesis. The replica/EVT lift $(\ln \ln K)/(2 \ln 2)$ was originally derived with Rice’s second-moment count. Equating the saddle’s mode count to the participation ratio $K_{\text{eff}}^{\text{PR}}$ is physically motivated—both are “number of independent modes”, and the witness shows the *maximum* sees PR-amplitude modes, not crossings—but it is *not derived from the saddle* (open hypothesis O1, Section 7). The witness confirms the mechanism; closing the saddle identification is a separate, still-open theoretical step.

Diagnose-and-fix history (full disclosure). The capacity run was not clean on the first attempt. (i) The initial α grid was placed too high (above the crossing), so every seed failed to solve and $P_{\text{solve}} \equiv 0$; the grid was lowered to $\{1.8, \dots, 2.8\}$ to bracket the half-crossing. (ii) A

`require_cuda` fail-fast guard (added so the GPU-priced job never silently runs a CPU smoke test) tripped on a host that was provisioned without a CUDA device, aborting non-zero by design; the job was resubmitted to a GPU host. (iii) A dead/unresponsive rental host required a further re-submit. None of these affected the reported numbers—they are data-pipeline incidents, not result corrections—but they are recorded here and in `EXPERIMENT.md` for full reproducibility.

What is not pinned. The exact $\alpha_c(\tau_s)$ curve at the corner $\tau_s \ll \tau$ is not pinned: it needs a fine time grid ($n_{\text{grid}} \approx 27,000$ per pattern at $\tau_s = 0.15$ ms) and, as noted, the capacity gap there is inside finite- N noise; we deliberately skipped that corner for the *capacity* test (it is exactly where the *witness* is cheapest and most decisive). The few-percent finite- T Whittle edge correction to $K_{\text{eff}}^{\text{PR}}$ (O2) and the white-afferent assumption (O3) are likewise not quantified here.

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